Modelo de Examen 2019

1) SVM

$$L_D = -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} y_i y_j \alpha_i \alpha_j x_i^T x_j + \sum_{i=1}^{N} \alpha_i$$

$$L_D = -\frac{1}{2}\sum_{i=1}^{4}\sum_{j=1}^{4} \alpha_i \alpha_j y_i y_j x_i^T x_j + \sum_{i=1}^{4} \alpha_i$$

$$\begin{pmatrix} -3 & 3 \end{pmatrix}\begin{pmatrix} -3 \\ 3 \end{pmatrix} \quad \begin{pmatrix} -3 & 3 \end{pmatrix}\begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

$$L_D = -\frac{1}{2}\sum_{i=1}^{4}\left[\alpha_i\alpha_1 y_i y_1 x_i^T x_1 + \alpha_i\alpha_2 y_i y_2 x_i^T x_2 + \alpha_i\alpha_3 y_i y_3 x_i^T x_3 + \alpha_i\alpha_4 x_i^T x_4\right] + \sum_{i=1}^{4} \alpha_i$$

$$L_D = -\frac{1}{2}\Big[\alpha_1\alpha_1 y_1 y_1 x_1^T x_1 + \alpha_1\alpha_2 y_1 y_2 x_1^T x_2 + \alpha_1\alpha_3 y_1 y_3 x_1^T x_3 + \alpha_1\alpha_4 x_1^T x_4 y_1 y_4 +$$
$$\alpha_2\alpha_1 y_2 y_1 x_2^T x_1 + \alpha_2\alpha_2 y_2 y_2 x_2^T x_2 + \alpha_2\alpha_3 y_2 y_3 x_2^T x_3 + \alpha_2\alpha_4 x_2^T x_4 y_2 y_4 +$$
$$\alpha_3\alpha_1 y_3 y_1 x_3^T x_1 + \alpha_3\alpha_2 y_3 y_2 x_3^T x_2 + \alpha_3\alpha_3 y_3 y_3 x_3^T x_3 + \alpha_3\alpha_4 x_3^T x_4 y_3 y_4 +$$
$$\alpha_4\alpha_1 y_4 y_1 x_4^T x_1 + \alpha_4\alpha_2 y_4 y_2 x_4^T x_2 + \alpha_4\alpha_3 y_4 y_3 x_4^T x_3 + \alpha_4\alpha_4 x_4^T x_4 y_4 y_4\Big]$$
$$+ \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$$

$$L_D = -\frac{1}{2}\Big[\alpha_1^2 \cdot 1\cdot 1 \cdot \begin{bmatrix}-3 & 3\end{bmatrix}\begin{bmatrix}-3\\3\end{bmatrix} + \alpha_1\alpha_2 \cdot 1\cdot 1 \begin{bmatrix}-3 & 3\end{bmatrix}\begin{bmatrix}0\\5\end{bmatrix} + \alpha_1\alpha_3 \cdot 1\cdot(-1)\begin{bmatrix}-3 & 3\end{bmatrix}\begin{bmatrix}1\\-3\end{bmatrix} + \alpha_1\alpha_4\begin{bmatrix}-3 & 3\end{bmatrix}\begin{bmatrix}2\\-4\end{bmatrix}1\cdot(-1) +$$
$$\alpha_2\alpha_1 \cdot 1\cdot 1\begin{bmatrix}0 & 5\end{bmatrix}\begin{bmatrix}-3\\3\end{bmatrix} + \alpha_2\alpha_2 \cdot 1\cdot 1\begin{bmatrix}0 & 5\end{bmatrix}\begin{bmatrix}0\\5\end{bmatrix} + \alpha_2\alpha_3 \cdot 1\cdot(-1)\begin{bmatrix}0 & 5\end{bmatrix}\begin{bmatrix}1\\-3\end{bmatrix} + \alpha_2\alpha_4\begin{bmatrix}0 & 5\end{bmatrix}\begin{bmatrix}2\\-4\end{bmatrix}1\cdot(-1) +$$
$$\alpha_3\alpha_1(-1)(1)\begin{bmatrix}1 & -3\end{bmatrix}\begin{bmatrix}-3\\3\end{bmatrix} + \alpha_3\alpha_2(-1)(1)\begin{bmatrix}1 & -3\end{bmatrix}\begin{bmatrix}0\\5\end{bmatrix} + \alpha_3\alpha_3(-1)(-1)\begin{bmatrix}1 & -3\end{bmatrix}\begin{bmatrix}1\\-3\end{bmatrix} + \alpha_3\alpha_4(-1)(-1)\begin{bmatrix}1 & -3\end{bmatrix}\begin{bmatrix}2\\-4\end{bmatrix} +$$
$$\alpha_4\alpha_1(-1)(1)\begin{bmatrix}2 & -4\end{bmatrix}\begin{bmatrix}-3\\3\end{bmatrix} + \alpha_4\alpha_2(-1)(1)\begin{bmatrix}2 & -4\end{bmatrix}\begin{bmatrix}0\\5\end{bmatrix} + \alpha_4\alpha_3(-1)(-1)\begin{bmatrix}2 & -4\end{bmatrix}\begin{bmatrix}1\\-3\end{bmatrix} + \alpha_4\alpha_4(-1)(-1)\begin{bmatrix}2 & -4\end{bmatrix}\begin{bmatrix}2\\-4\end{bmatrix}\Big]$$
$$+ \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$$

$$L_D = -\frac{1}{2}\Big[18\alpha_1^2 + 15\alpha_1\alpha_2 + 18\alpha_1\alpha_3 + 18\alpha_1\alpha_4 +$$
$$15\alpha_1\alpha_2 + 25\alpha_2^2 + 15\alpha_2\alpha_3 + 20\alpha_2\alpha_4 +$$
$$12\alpha_3\alpha_1 + 15\alpha_3\alpha_2 + 10\alpha_3^2 + 14\alpha_3\alpha_4 +$$
$$18\alpha_4\alpha_1 + 20\alpha_4\alpha_2 + 14\alpha_4\alpha_3 + 20\alpha_4^2\Big] + \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$$

$$L_D = -\frac{1}{2}\Big[18\alpha_1^2 + 24\alpha_1\alpha_3 + 36\alpha_1\alpha_4 + 25\alpha_2^2 + 30\alpha_2\alpha_3 + 40\alpha_2\alpha_4 + 10\alpha_3^2 + 30\alpha_1\alpha_2 + 28\alpha_3\alpha_4 + 20\alpha_4^2\Big] + \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$$

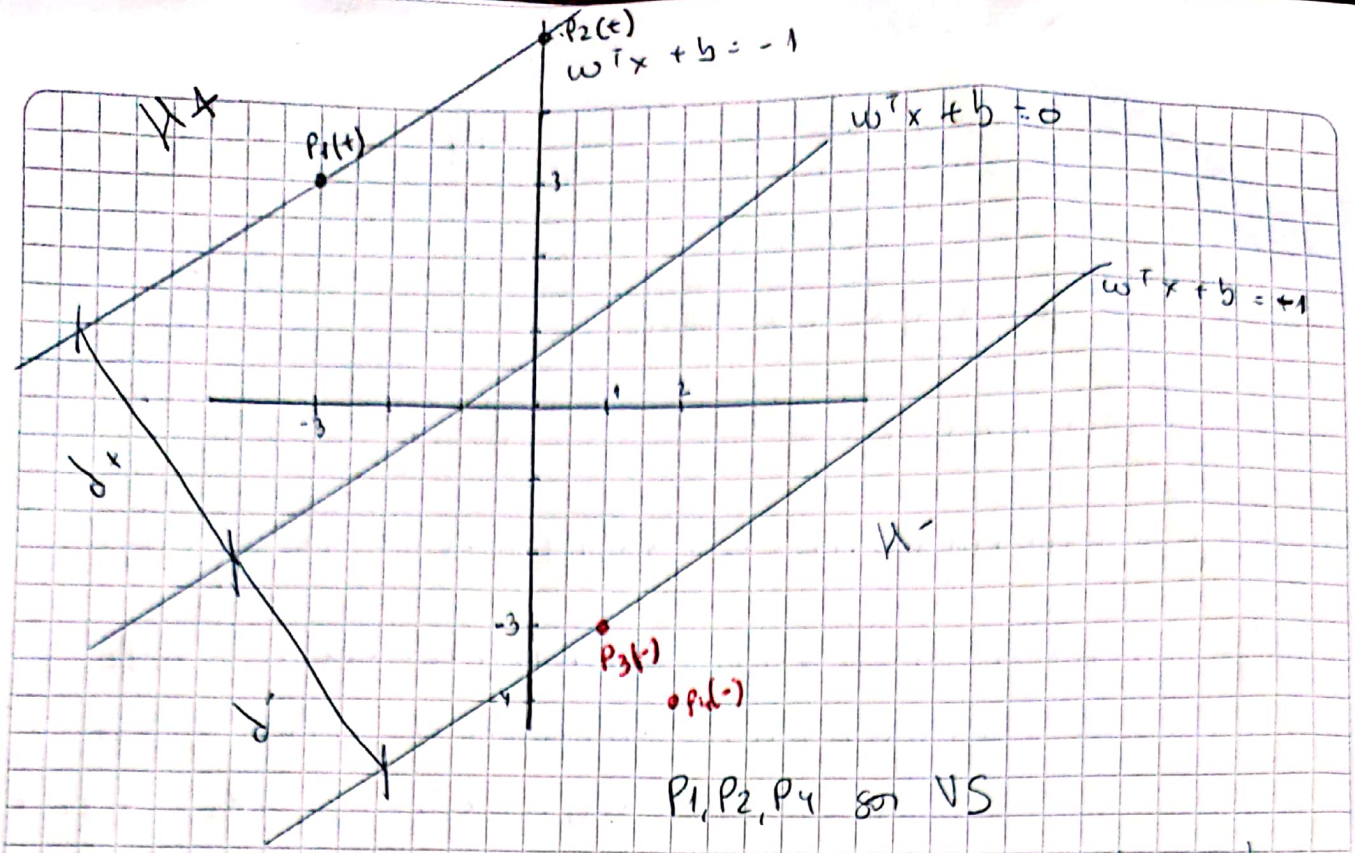
Sujeto a: $\sum_{i=1}^{n} y_i\alpha_i = 0 = y_1\alpha_1 + y_2\alpha_2 + y_3\alpha_3 + y_4\alpha_4 = 0$

$$= 1\alpha_1 + 1\alpha_2 - 1\alpha_3 - 1\alpha_4 = 0$$

$\alpha_i \geq 0 \rightarrow \alpha_1 \geq 0 \quad \alpha_2 \geq 0 \quad \alpha_3 \geq 0 \quad \alpha_4 \geq 0$

Notas: $\max L_D = -9\alpha_1^2 - 12\alpha_1\alpha_3 - 15\alpha_1\alpha_2 - 18\alpha_1\alpha_4 - 25\alpha_2^2 - 15\alpha_3\alpha_2 - 20\alpha_2\alpha_4 - 5\alpha_3^2 - 14\alpha_3\alpha_4 - 10\alpha_4^2 + \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$

Escaneado con CamScanner



$$d_+ = \frac{1}{\|w\|} = \frac{1}{\sqrt{w^T \cdot w}} =$$

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3) Error $S_i = \frac{(Sd_i - So_i)^2}{2}$; Error patron $= \sum_{i=1}^{n}$ Error S_i ; Error Global $= \frac{1}{L} \sum_{i=1}^{L}$ Error Patron i ;

Error $P_1 = \frac{(0,02)^2}{2} + \frac{(0,11)^2}{2} = 6,25 \times 10^{-3}$

Error $P_2 = \frac{(0,12)^2}{2} - \frac{(0,02)^2}{2} = 7,4 \times 10^{-3}$

Error $P_3 = \frac{(0,03)^2}{2} + \frac{(0,16)^2}{2} = 1,325 \times 10^{-2}$

Error $P_4 = \frac{(0,14)^2}{2} + \frac{(0,12)^2}{2} = 0,017$

Error $P_5 = \frac{(0,04)^2}{2} + \frac{(0,09)^2}{2} = 4,85 \times 10^{-3}$

Error $P_6 = \frac{(0,12)^2}{2} + \frac{(0,07)^2}{2} = 1,04 \times 10^{-2}$

Error $P_7 = \frac{(0,13)^2}{2} + \frac{(0,17)^2}{2} = 2,29 \times 10^{-2}$

Error $P_8 = \frac{(0,04)^2}{2} + \frac{(0,17)^2}{2} = 1,525 \times 10^{-2}$

Error $P_9 = \frac{(0,04)^2}{2} + \frac{(0,06)^2}{2} = 2,6 \times 10^{-3}$

Error $P_{10} = \frac{(0,04)^2}{2} + \frac{(0,06)^2}{2} = 2,6 \times 10^{-3}$

Error Global $= \frac{0,10025}{10} = 0,010025$

$= 0,01025$

"No, el entrenamiento no debería detenerse debido a que el error global (0,010125) es mayor al mínimo aceptable (0,01) debe seguir iterando hasta alcanzar un error global aceptablemente pequeño."

Notas:

Escaneado con CamScanner

Dada la evolución de peso w_{H1-S1}, calcule el término de morrendo asociado a que se utiliza en $t=4$

$$w_{ij}(t+1) = w_{ij}(t) + \alpha \cdot \delta_{pj} \cdot y_{pi} + \beta(w_{ij}(t) - w_{ij}(t-1))$$

$$w_{H1-S1}(2) = w_{H1-S1}(1) + 0,2 \cdot 0,09 \cdot 0,99 + 0,9(w_{H1-S1}(1) - w_{H1-S1}(0))$$

$$= -0,28 + 0,2 \cdot 0,09 \cdot 0,99 + 0,9(-0,28 + 0,28)$$

$$w_{H1S1}(5) = w_{H1S1}(4) + \alpha \times \delta_{H1} \times S_1 + \beta(w_{H1S1}(4) - w_{H1S1}(3))$$

$$= -0,28 + 0,2 \times 0,09 \times 0,99 + 0,9(-0,28 + 0,12)$$

$$= -0,01444$$

	x1	x2
A	19,2	55,2
B	28,7	74,9
C	35,7	46,1
D	44,7	55,8
E	47,0	64,6

$$dist_{ij} = \sqrt{(x_{1i} - x_{1j})^2 + (x_{2i} - x_{2j})^2}$$

$dist_{AB} = \sqrt{(19,2 - 28,7)^2 + (55,2 - 74,9)^2} = 21,87 \rightarrow A$

$dist_{AC} = \sqrt{(19,2 - 35,7)^2 + (55,2 - 46,1)^2} = 18,84 \rightarrow B$

$dist_{AD} = \sqrt{(19,2 - 44,7)^2 + (55,2 - 55,8)^2} = 25,51 \rightarrow C$

$dist_{AE} = \sqrt{(19,2 - 47,0)^2 + (55,2 - 64,6)^2} = 29,35 \rightarrow D$

$dist_{BC} = \sqrt{(28,7 - 35,7)^2 + (74,9 - 46,1)^2} = 29,64 \rightarrow E$

$dist_{BD} = \sqrt{(28,7 - 44,7)^2 + (74,9 - 55,8)^2} = 24,92 \rightarrow F$

$dist_{BE} = \sqrt{(28,7 - 47,0)^2 + (74,9 - 64,6)^2} = 21$

$dist_{CD} = \sqrt{(35,7 - 44,7)^2 + (46,1 - 55,8)^2} = 13,23 \rightarrow x$

$dist_{CE} = \sqrt{(35,7 - 47,0)^2 + (46,1 - 64,6)^2} = 21,68 \rightarrow y$

$dist_{DE} = \sqrt{(44,7 - 47,0)^2 + (55,8 - 64,6)^2} = 9,10 \rightarrow$

Single Link

	A	B	C	D	E
A	0				
B	21,87	0			
C	18,84	29,64	0		
D	25,51	24,92	13,23	0	
E	29,35	21	21,68	9,10	0

$G1 = D + E$

	A	B	C	G1
A	0			
B	21,87	0		
C	18,84	29,64	0	
G1	25,51	21	13,23	0

$G2 = C + G1$

	A	B	G2
A	0		
B	21,87	0	
G2	18,84	21	0

$G3 = B + G2$

	A	G3
A	0	
G3	18,84	0

Notas:

Escaneado con CamScanner

	A	B	C	D	E
A	∞				
B	21,87	∞			
C	18,84	29,64	∞		
D	25,51	24,92	13,23	∞	
E	29,35	21	21,68	9,10	∞

$G1 = D + E$

	A	B	C	G1
A	∞			
B	21,87	∞		
C	18,84	29,64	∞	
G1	29,35	24,92	21,68	∞

$G2 = A + C$

	B	G1	G2
B	∞		
G1	21,92	∞	
G2	29,64	29,35	∞

$G3 = B + G1$

	G2	G3
G2	∞	
G3	69,64	∞

4) Primero en profundidad

$t_0 = (A)$

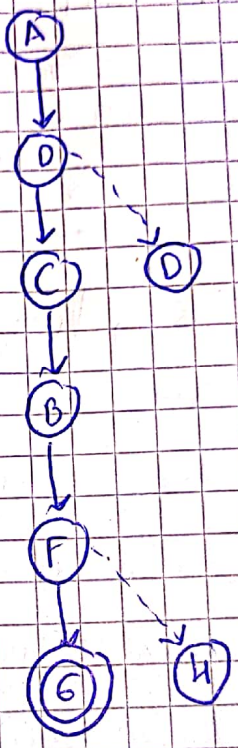
$t_1 = (D^A)\ E^A$

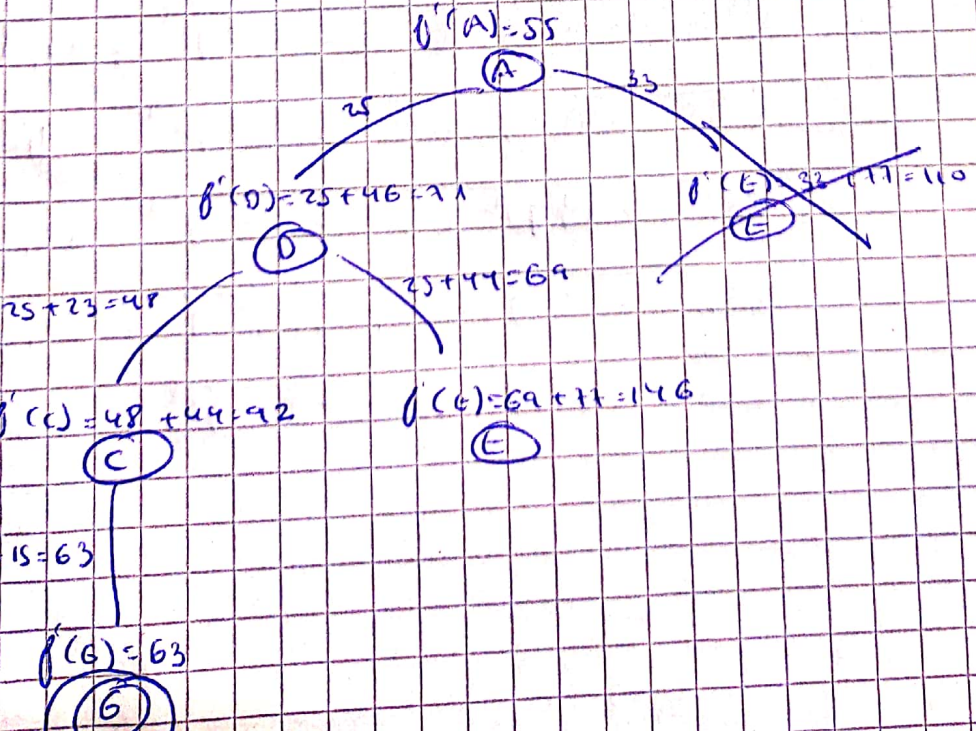
$t_2 = (C^D)\ E^D\ E^A$

$t_3 = (B^C)\ G^C\ E^D$

$t_4 = (F^B)\ G^C\ E^D$

$t_5 = (G^F)\ H^F\ \cancel{G^C}\ E^D$





Escaneado con CamScanner